Convergence Simulations Model C

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1 Model Structure

Model C is Model B (hierarchical over sessions) with the distractor entering the psychometric function as a covariate. Instead of fitting unimanual and bimanual trials as separate groups, it adds a term $\beta_2 d$ for the distractor amplitude d, so the groups are the two hands.

1.1 Data and indexing

The data are N binary trials. Trial i has

- a response $y_i \in \{0, 1\}$, the monkey's categorization of the stimulus as "low" (0) or "high" (1);
- a standardized stimulus amplitude x_i. As in Models A and B, the raw amplitudes are centered and scaled by the mean and standard deviation of the distinct amplitude levels;
- a distractor covariate d_i: the distractor amplitude divided by the same standard deviation (14.63) but not centered, so that $d_i = 0$ on unimanual trials (no distractor). On bimanual trials d_i ranges from 0.41 (amplitude 6) to 2.19 (amplitude 32). The covariate row is $(1, x_i, d_i)$;
- a hand $j[i] \in \{\text{left}, \text{right}\}$;
- a session $s[i] \in \{1, \ldots, S\}$; the real data have $S = 42$.

Every session uses a single distractor amplitude on its bimanual trials, so within a session d_i only switches between 0 and that session's value d_s. The real sessions use 11 different amplitudes, most often 32 (13 sessions), 6 (8 sessions), 24 (7 sessions) and 12 (6 sessions).

1.2 Likelihood

Each response is Bernoulli,

$$y_i \sim \text{Bernoulli}(\psi_i), \qquad \psi_i = \psi\big(x_i, d_i;\ \theta_{j[i],s[i]}\big),$$

with the psychometric function

$$\psi(x, d; \theta) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x + \beta_2 d), \qquad \theta = (\beta_0, \beta_1, \beta_2, \gamma_h, \gamma_l).$$

As in Model B, γ_h is the probability of responding "high" at very low amplitudes and γ_l the probability of responding "low" at very high amplitudes. β_2 shifts the curve on bimanual trials in proportion to the distractor amplitude: a positive β_2 makes "high" responses more likely, moving the PSE to lower amplitudes. The lapse rates are shared by the unimanual and bimanual trials of a hand and session. In the code ψ_i is clipped to $[10^{-8}, 1 - 10^{-8}]$ before entering the Bernoulli likelihood, for numerical stability.

1.3 Session-level

For each hand j and session s, the five parameters are drawn independently from hand-specific distributions:

$$\begin{aligned}
\beta_{k,js} &\sim \mathcal{N}\big(\mu_{\beta_k,j},\ \sigma_{\beta_k,j}\big), \qquad k = 0, 1, 2, \\
\gamma_{h,js} &= 0.25\, w_{h,js}, \qquad w_{h,js} \sim \text{LogitNormal}\big(\mu_{\gamma_h,j},\ \sigma_{\gamma_h,j}\big), \\
\gamma_{l,js} &= 0.25\, w_{l,js}, \qquad w_{l,js} \sim \text{LogitNormal}\big(\mu_{\gamma_l,j},\ \sigma_{\gamma_l,j}\big).
\end{aligned}$$

The two hands are independent of each other (no pooling across hands).

1.4 Hand-level Hyperpriors

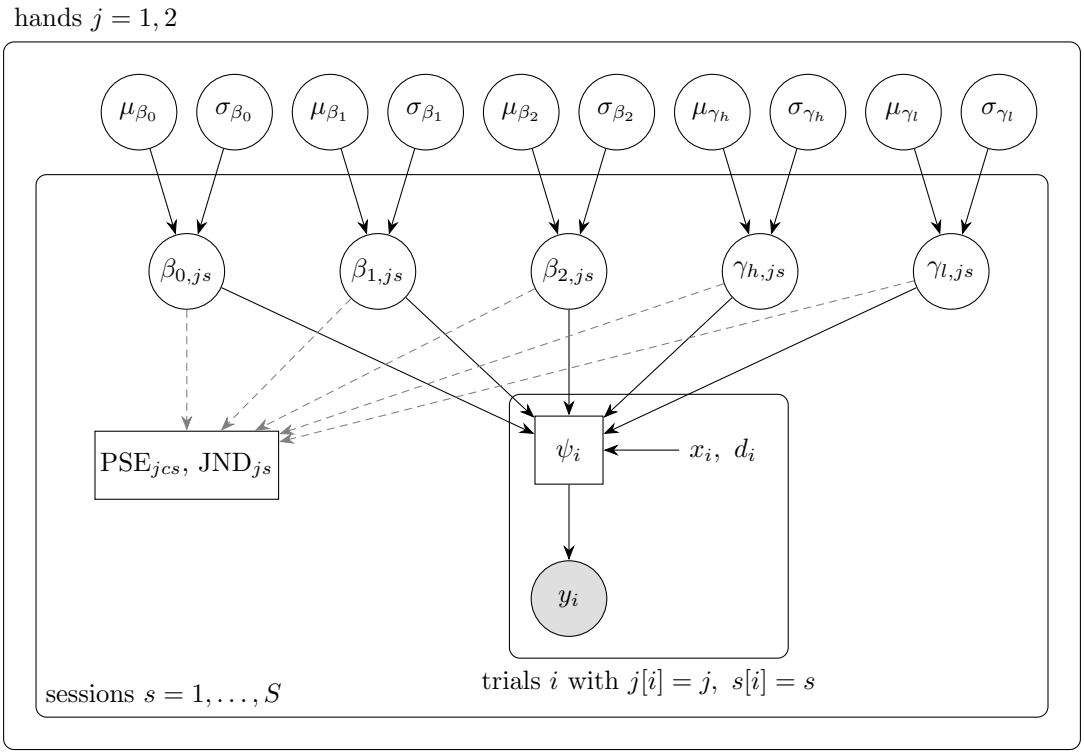
Each hand has 10 hyperparameters. Those of β_0, β_1 and the lapse rates have the same priors as in Model B:

$$\begin{aligned}
\mu_{\beta_0,j} &\sim \mathcal{N}(0,\ 2), & \sigma_{\beta_0,j} &\sim \text{HalfNormal}(2), \\
\mu_{\beta_1,j} &\sim \text{LogNormal}(\log 4,\ 0.5), & \sigma_{\beta_1,j} &\sim \text{Exponential}(\lambda = 1), \\
\mu_{\beta_2,j} &\sim \mathcal{N}(0,\ 1), & \sigma_{\beta_2,j} &\sim \text{HalfNormal}(1), \\
\mu_{\gamma_h,j},\ \mu_{\gamma_l,j} &\sim \mathcal{N}(-3,\ 1.5), & \sigma_{\gamma_h,j},\ \sigma_{\gamma_l,j} &\sim \text{HalfNormal}(0.5).
\end{aligned}$$

All hyperparameters are independent. The Model B report explains the priors shared with Model B. The priors of β_2 were chosen from what the parameter means, not from fits to the real data:

- μ_{β_2}: centered at 0 because the distractor could push responses either way. With d at most 2.19, the typical shift $\mu_{\beta_2} d$ at the largest distractor lies within ± 4.3 logits with 95% prior probability, the same scale as the prior on μ_{β_0}.
- σ_{β_2}: median 0.67, 95% point 1.96, allowing sessions to differ in their distractor effect by about as much as the hand-level effect itself.

1.5 Graphical Model



Circles are random variables (shaded: observed), rectangles are deterministic functions, and x_i, d_i are fixed covariates. Dashed arrows lead to derived quantities that do not feed back into the likelihood.

1.6 Derived Quantities: PSE and JND

The PSE now depends on the distractor. For each hand and session it is recorded for both conditions c: unimanual ($d = 0$) and bimanual ($d = d_s$, the session's distractor). Solving $\psi = \frac{1}{2}$ for x gives

$$\text{PSE}_c = \frac{1}{\beta_1}\left[-\big(\beta_0 + \beta_2 d\big) + \log\frac{1 - 2\gamma_h}{1 - 2\gamma_l}\right], \qquad d = 0 \text{ (uni)},\ d_s \text{ (bi)}.$$

The JND does not depend on the distractor, since $\beta_2 d$ only shifts the curve:

$$\text{JND} = \frac{1}{2\beta_1}\,\log\frac{(3 - 4\gamma_h)(3 - 4\gamma_l)}{(1 - 4\gamma_h)(1 - 4\gamma_l)}.$$

Both are in units of the standardized stimulus.

1.7 Implementation

As in Model B, the distributions are written as deterministic transforms of standard random variables (non-centered parameterizations). Unlike Model B, the HalfNormal sds σ_{β_0}, σ_{β_2}, σ_{γ_h} and σ_{γ_l} are scaled HalfNormal(1) variables rather than absolute values of standard normals: with the absolute value the sign is not identified, chains settle on opposite signs, and the sampler diagnostics report a failure that does not affect the model. The two forms define the same prior.

2 Synthetic Data Generation

The synthetic data follow Model C with true hyperparameters derived from Model B's fit to the real data (posterior means in `Results_B.pkl`), so that they resemble the real data. These values only generate data; the priors do not use them. Model B has no β_2 and fits unimanual and bimanual trials separately, so for each hand (`SynthData_C.py`):

- $\mu_{\beta_0}, \sigma_{\beta_0}$: Model B's unimanual hyperparameters (β_0 at $d = 0$);
- $\mu_{\beta_1}, \sigma_{\beta_1}$: the average of Model B's unimanual and bimanual hyperparameters;
- μ_{β_2}: the slope through the origin of each session's bimanual shift in β_0 (bimanual minus unimanual posterior mean) on that session's d_s, which is what $\beta_2 d$ describes in Model C;
- σ_{β_2}: the sd across sessions of that shift divided by d_s;
- $\mu_{\gamma_h}, \mu_{\gamma_l}$: the logit of 4 times the mean lapse rate over all unimanual and bimanual session estimates, since Model C shares lapse rates between them;
- $\sigma_{\gamma_h}, \sigma_{\gamma_l}$: the average of Model B's unimanual and bimanual hyperparameters.

Table 1: True hyperparameters. Lapse-rate hyperparameters are on the logit scale; the typical lapse rate is 0.25 $\text{logit}^{-1}(\mu_\gamma)$.

	left	right
μ_{β_0} / σ_{β_0}	−1.487 / 0.763	−1.056 / 1.011
μ_{β_1} / σ_{β_1}	4.644 / 0.776	4.460 / 0.394
μ_{β_2} / σ_{β_2}	0.933 / 0.622	1.129 / 0.831
μ_{γ_h} / σ_{γ_h}	−2.802 / 0.475	−2.414 / 0.366
μ_{γ_l} / σ_{γ_l}	−3.197 / 0.383	−3.204 / 0.406
typical γ_h / γ_l	0.0143 / 0.0098	0.0205 / 0.0098

Unlike the earlier hand-picked values for Model B's study, the between-session sds of the slope are substantial (0.78 and 0.39), and those of the distractor coefficient (0.62 and 0.83) are close to those of β_0 (0.76 and 1.01).

As in Model B's study, the model was fit to 10 dataset conditions in two sweeps, each run twice with new data:

- the sessions sweep varies the number of sessions (5, 10, 20, 40 and 80, each with 308 trials), mainly testing the hyperparameters;
- the trials sweep varies trials per session (20, 40, 80, 160 and 320, with 42 sessions), mainly testing the session-level parameters.

Every replicate draws new session-level parameters from the true hyperparameters. Each simulated session gets one distractor amplitude, drawn from the real sessions' amplitudes, and its trials cycle through the four cells hand × (unimanual, bimanual), with each stimulus resampled from that cell's stimuli in the real design. Within a replicate the datasets of a sweep are nested (the first n sessions, or the first t trials of every session). The real data have 42 sessions with about 308 trials each, so the 40-session and 320-trial fits are the most representative.

3 Model Fit Results

Every figure and number in this report is produced by `Convergence_C.py`; the cell "Numbers for Convergence_C_Report.tex" prints the numbers that the earlier cells do not, and also runs on the saved results without refitting. Hyperparameter entries are averaged over the two replicates; session-level RMSEs and coverages are pooled over them (84 sessions per hand in the trials sweep). Coverages of the hyperparameters are over 4 fits per size (2 hands $\times$ 2 replicates).

3.1 Sampling Diagnostics

The 20 fits had 2 divergences in total, one each in a 5-session and a 10-session fit, too few to bias the results. $\hat{R} \leq 1.008$ and the bulk ESS is at least 532 for every hyperparameter, and $\hat{R} \leq 1.007$ for every session-level parameter. The whole study takes about 0.7 hours.

Table 2: Sampler diagnostics over both replicates. $\hat{R}$ and bulk ESS are taken over the hyperparameters.

Sweep	Size	max $\hat{R}$	min ESS	divergences
sessions	5	1.005	1467	1
sessions	10	1.004	1194	1
sessions	20	1.004	1447	0
sessions	40	1.003	1524	0
sessions	80	1.003	1309	0
trials	20	1.007	598	0
trials	40	1.008	532	0
trials	80	1.008	553	0
trials	160	1.006	794	0
trials	320	1.005	847	0

3.2 Hyperparameter means

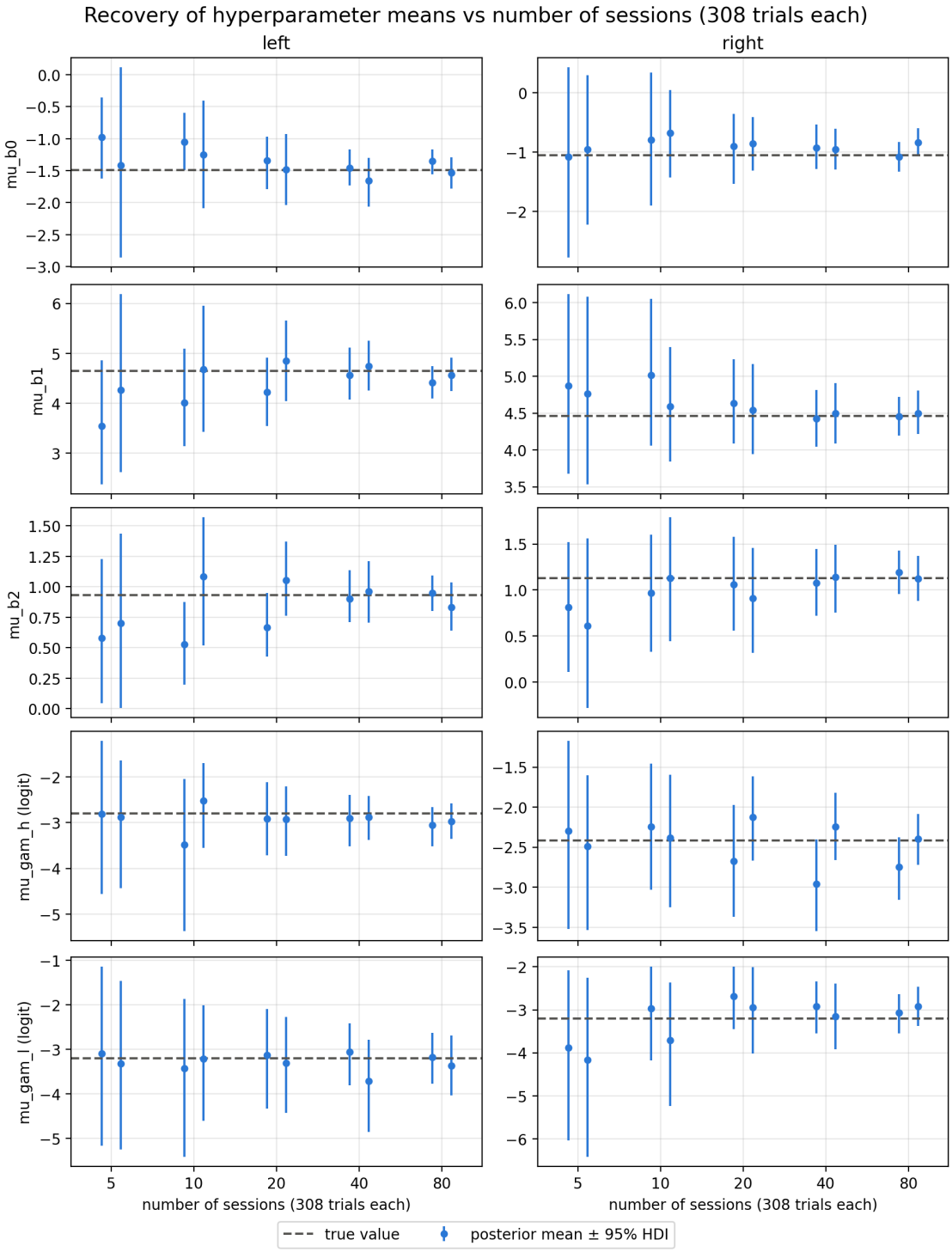


Figure 1: Posterior mean and 95% HDI of the hand-level means against the number of sessions (308 trials each). The two replicates are drawn side by side; the dashed line is the true value.

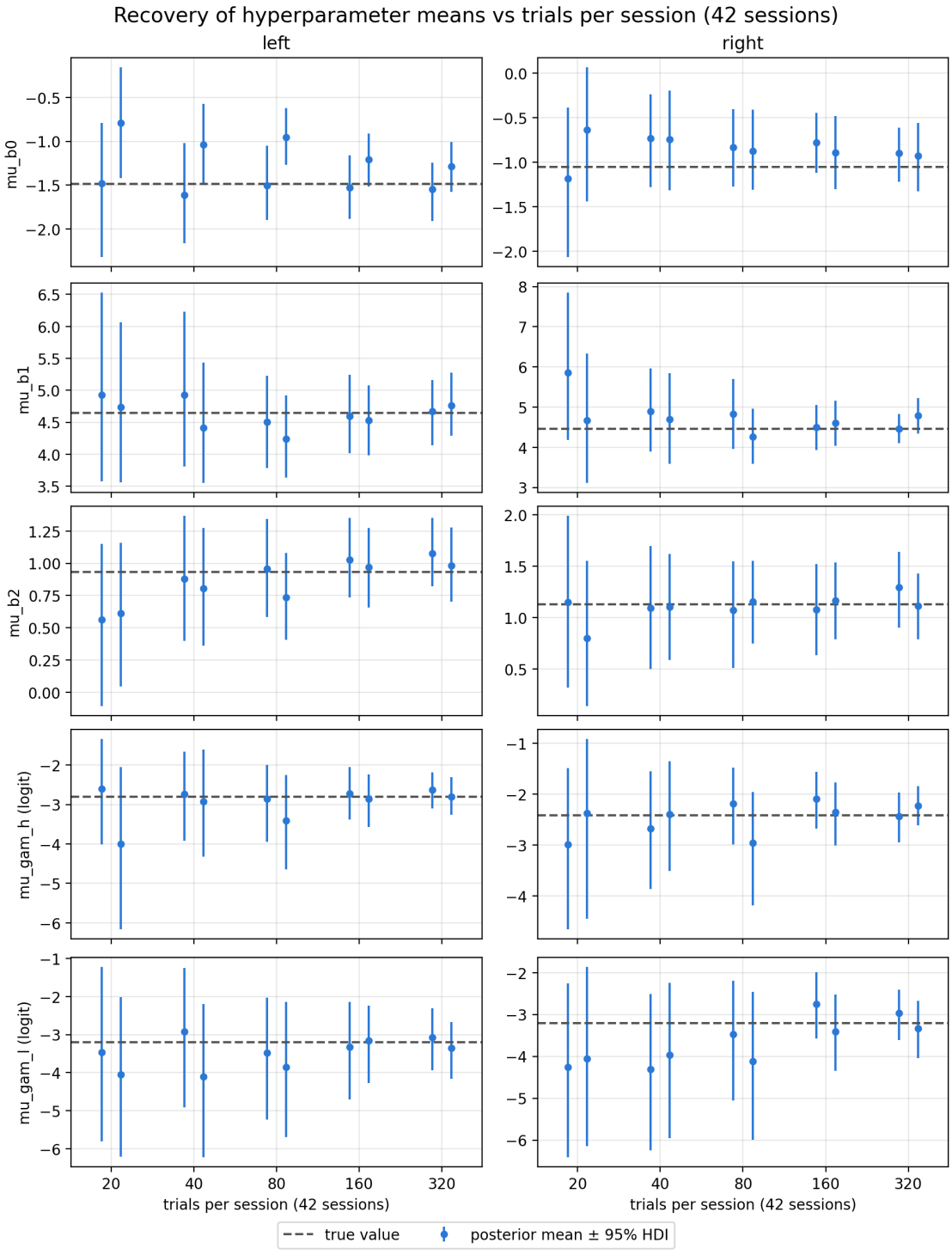


Figure 2: As Figure 1, against trials per session (42 sessions).

Table 3: Hand-level means at 40 sessions (sessions sweep): true value and posterior mean [95% HDI], averaged over replicates. Lapse-rate means are on the logit scale.

	left		right	
	true	posterior	true	posterior
μ_{β_0}	-1.49	-1.56 [-1.90, -1.24]	-1.06	-0.94 [-1.29, -0.57]
μ_{β_1}	4.64	4.65 [4.16, 5.18]	4.46	4.46 [4.06, 4.86]
μ_{β_2}	0.93	0.93 [0.71, 1.18]	1.13	1.11 [0.74, 1.47]
μ_{γ_h}	-2.80	-2.89 [-3.45, -2.41]	-2.41	-2.60 [-3.10, -2.11]
μ_{γ_l}	-3.20	-3.39 [-4.34, -2.60]	-3.20	-3.04 [-3.73, -2.37]

All five hand-level means are recovered without visible bias, and in the sessions sweep their HDIs contain the true value in every fit at 40 and 80 sessions (Table 3). In the trials sweep, μ_{β_0} misses the truth in one of the four fits at each of 20, 40 and 80 trials. The HDIs narrow steadily with the number of sessions; averaged over hands and replicates, the width for μ_{β_0} falls from 2.49 at 5 sessions to 0.69 at 40 and 0.46 at 80.

Distractor effect μ_{β_2}. The new parameter is the best-determined mean relative to its size: at 40 sessions the HDIs are 0.47 (left) and 0.73 (right) wide, around true values of 0.93 and 1.13, so both exclude 0. At 5 sessions the HDI width is still 1.46 and the HDIs reach 0. In the trials sweep, where the number of sessions is fixed at 42, the width only falls from 1.36 at 20 trials to 0.62 at 320: like the other means, μ_{β_2} is limited by the number of sessions more than by trials per session.

Slope μ_{β_1}. Recovered well for both hands, in contrast with Model B's study, where a true slope of 6 for the right hand was poorly identified. The true slopes here, about 4.5, give a rise from $\psi = 0.25$ to 0.75 wide enough to span several stimulus levels.

Lapse-rate means. μ_{γ_h} and μ_{γ_l} are recovered, but they are the least precise means: at 80 sessions their HDIs are still 0.71–0.82 (μ_{γ_h}) and 0.91–1.25 (μ_{γ_l}) wide on the logit scale. Lapses are rare and only show at the stimulus levels where the sigmoid has reached its floor or ceiling.

3.3 Hyperparameter standard deviations

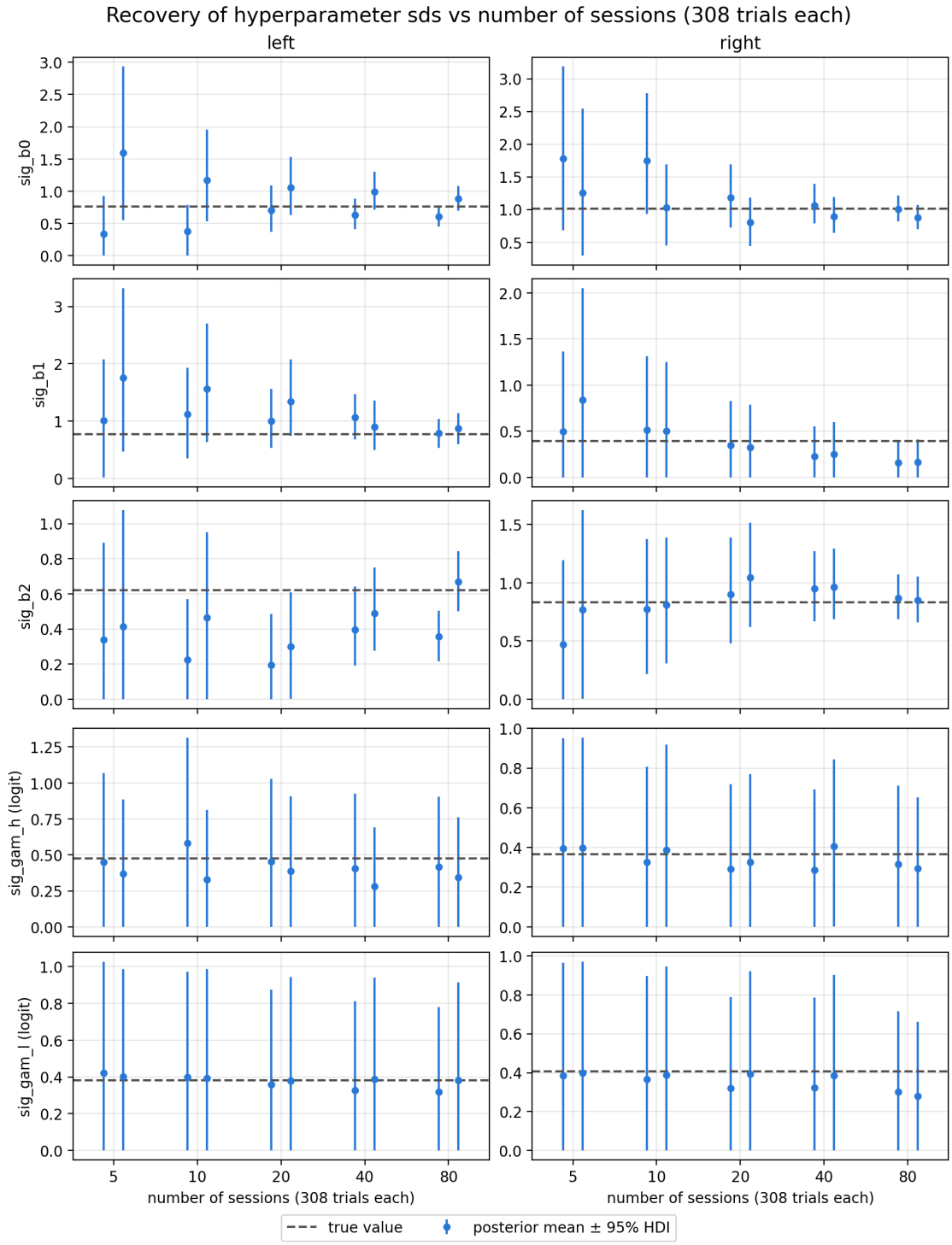


Figure 3: Posterior mean and 95% HDI of the between-session standard deviations against the number of sessions.

Table 4: Between-session sds at 80 sessions (sessions sweep): true value and posterior mean [95% HDI], averaged over replicates, and the fraction of the two replicates whose HDI contains the true value.

	left			right		
	true	posterior	cov.	true	posterior	cov.
σ_{β_0}	0.76	0.74 [0.58, 0.92]	1.0	1.01	0.94 [0.76, 1.14]	1.0
σ_{β_1}	0.78	0.83 [0.57, 1.09]	1.0	0.39	0.17 [0.00, 0.40]	0.5
σ_{β_2}	0.62	0.51 [0.36, 0.67]	0.5	0.83	0.86 [0.68, 1.06]	1.0
σ_{γ_h}	0.48	0.38 [0.00, 0.83]	1.0	0.37	0.31 [0.00, 0.68]	1.0
σ_{γ_l}	0.38	0.35 [0.00, 0.85]	1.0	0.41	0.29 [0.00, 0.69]	1.0

σ_{β_0}, σ_{β_1}, σ_{β_2}. The between-session sds of the three logistic coefficients are identified: their HDIs narrow with the number of sessions and, at 80 sessions, are 0.32–0.52 wide. Two caveats (Table 4):

- For the left hand, one of the two replicates misses the true σ_{β_2} at 80 sessions. In Figure 3 the left-hand σ_{β_2} estimates lie below the truth at most sizes in both replicates. Whether this is a bias or the particular draws of session effects cannot be told from two replicates.
- For the right hand, σ_{β_1} drifts down as sessions are added, to 0.17 at 80 sessions against a true 0.39. Both replicates' HDIs end near the truth, and one of them just misses it. The smallest true sd of the three is the hardest to separate from zero.

In the trials sweep, σ_{β_0} misses the truth in two of the four fits at 40 trials, one at 80, and none at 160 and 320.

Lapse-rate sds σ_{γ_h}, σ_{γ_l}. Not identified at these data sizes. Their HDIs always start at 0 and are still 0.68–0.85 wide at 80 sessions; averaged over hands, the width shrinks only from about 1.0 at 5 sessions to 0.76–0.77 at 80, and the posterior means barely move. Their true values (0.37–0.48) happen to be close to the prior median (0.34), so the 100% coverage says little: the posteriors largely reflect the prior.

3.4 Session-level parameters

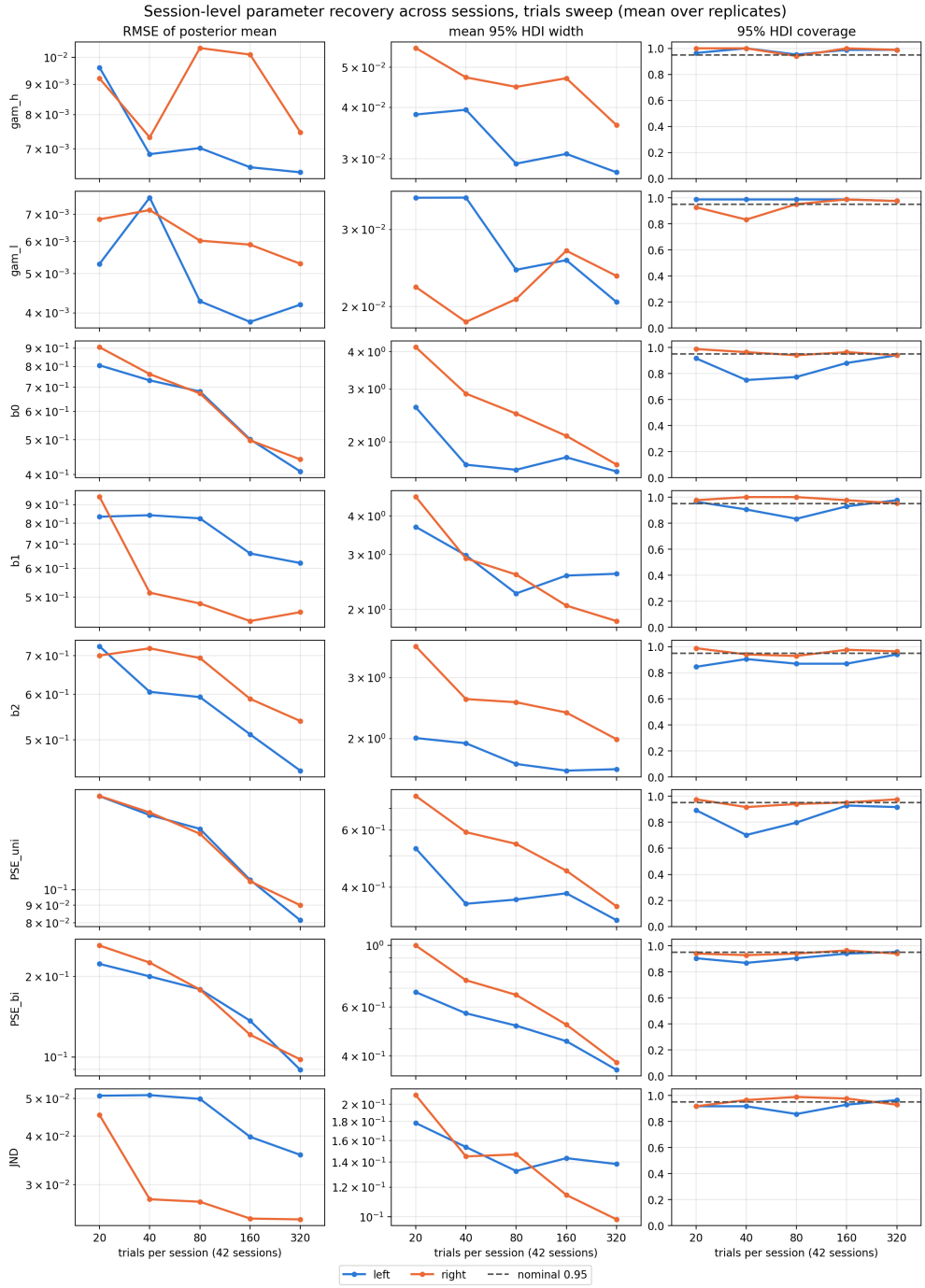


Figure 4: Session-level recovery against trials per session (42 sessions), averaged over replicates. Left: RMSE of the posterior mean across sessions. Middle: mean 95% HDI width. Right: fraction of sessions whose HDI contains the true value.

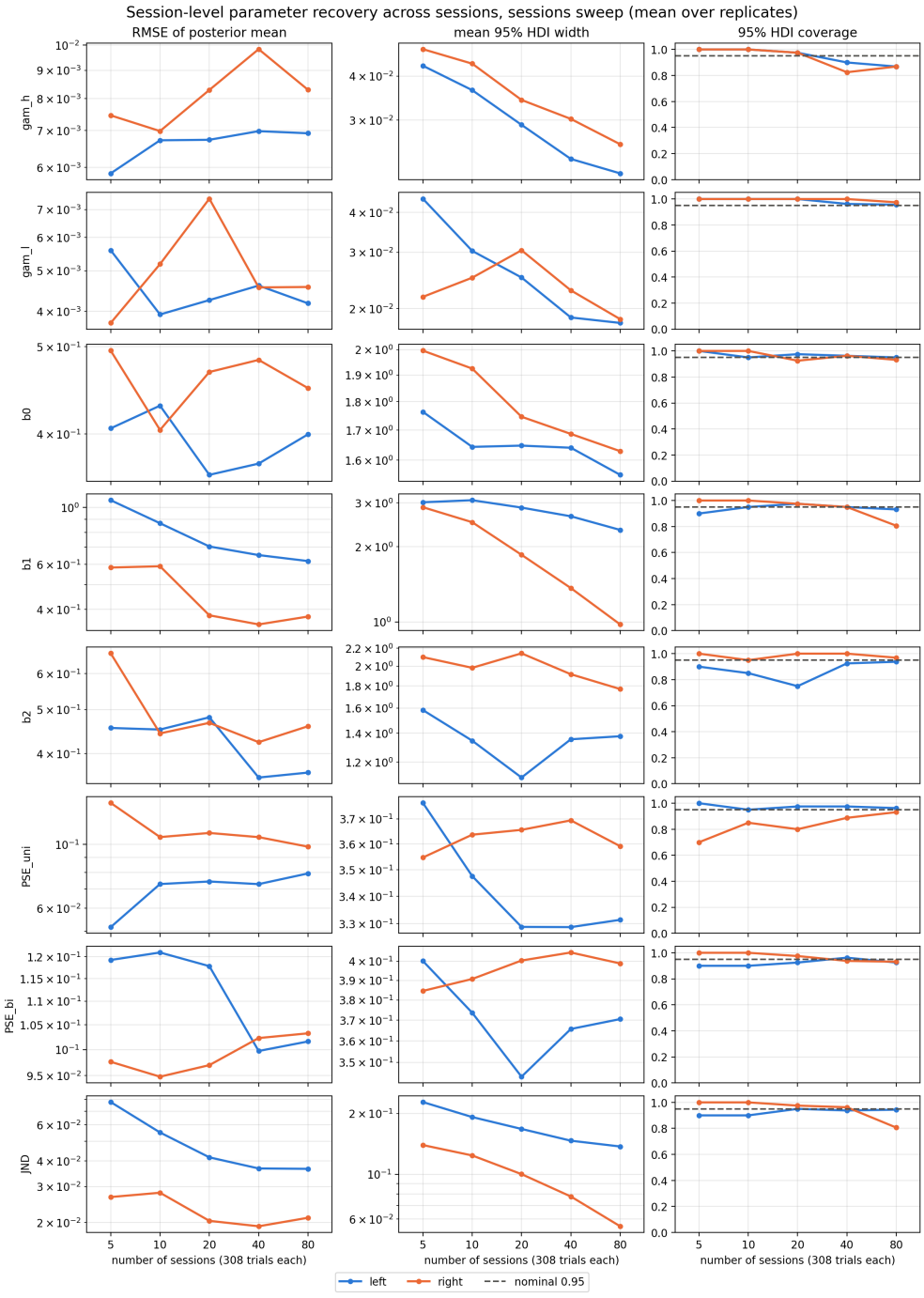


Figure 5: As Figure 4, against the number of sessions (308 trials each).

Table 5: Session-level recovery at 320 trials per session (42 sessions, both replicates): RMSE of the posterior mean / mean HDI width / coverage.

	left	right
PSE, unimanual	0.083 / 0.32 / 0.92	0.090 / 0.35 / 0.98
PSE, bimanual	0.090 / 0.36 / 0.95	0.098 / 0.38 / 0.94
JND	0.036 / 0.14 / 0.96	0.024 / 0.10 / 0.93
β_0	0.42 / 1.59 / 0.94	0.44 / 1.68 / 0.94
β_1	0.63 / 2.60 / 0.98	0.46 / 1.83 / 0.95
β_2	0.45 / 1.63 / 0.94	0.54 / 1.99 / 0.96
γ_h	0.006 / 0.028 / 0.99	0.007 / 0.036 / 0.99
γ_l	0.004 / 0.020 / 0.98	0.005 / 0.023 / 0.98

PSE and JND. The per-session PSEs are recovered well in both conditions: at 320 trials the RMSE is 0.08–0.10 standardized units, with coverage 0.92–0.98 (Table 5). The bimanual PSE is only slightly less precise than the unimanual one, although it also depends on β_2. The JND RMSE is 0.024–0.036. Error and HDI width fall steadily with trials per session (Figure 4).

β_0, β_1, β_2. The individual coefficients are much less precise than the PSEs computed from them (HDI widths 1.6–2.6 at 320 trials, against 0.32–0.38 for the PSEs): the data constrain the point where the curve crosses 0.5 better than the separate coefficients that place it there. Coverage is close to nominal at 320 trials (0.94–0.98). At 40 and 80 trials it is lower for β_0 (0.86 at both, averaged over hands) and the unimanual PSE (0.81 and 0.87).

Session-level β_2 and the distractor amplitude. Each session's β_2 is informed only by the shift between its unimanual and bimanual trials at its single distractor amplitude, and that shift is $\beta_2 d_s$. A small d_s therefore leaves β_2 poorly determined (Figure 6, Table 6): at 320 trials the HDI is 2.6 wide for sessions with a 6 μm distractor and 1.2 for 32 μm, and the correlation between d_s and HDI width is -0.88. Poorly determined sessions are pulled toward the hand mean, so across sessions the posterior means vary less than the true values: regressed on the true β_2, they have a slope of 0.43 (left) and 0.58 (right) instead of 1. Coverage stays near nominal overall (0.94 and 0.96), so the HDIs reflect this uncertainty correctly.

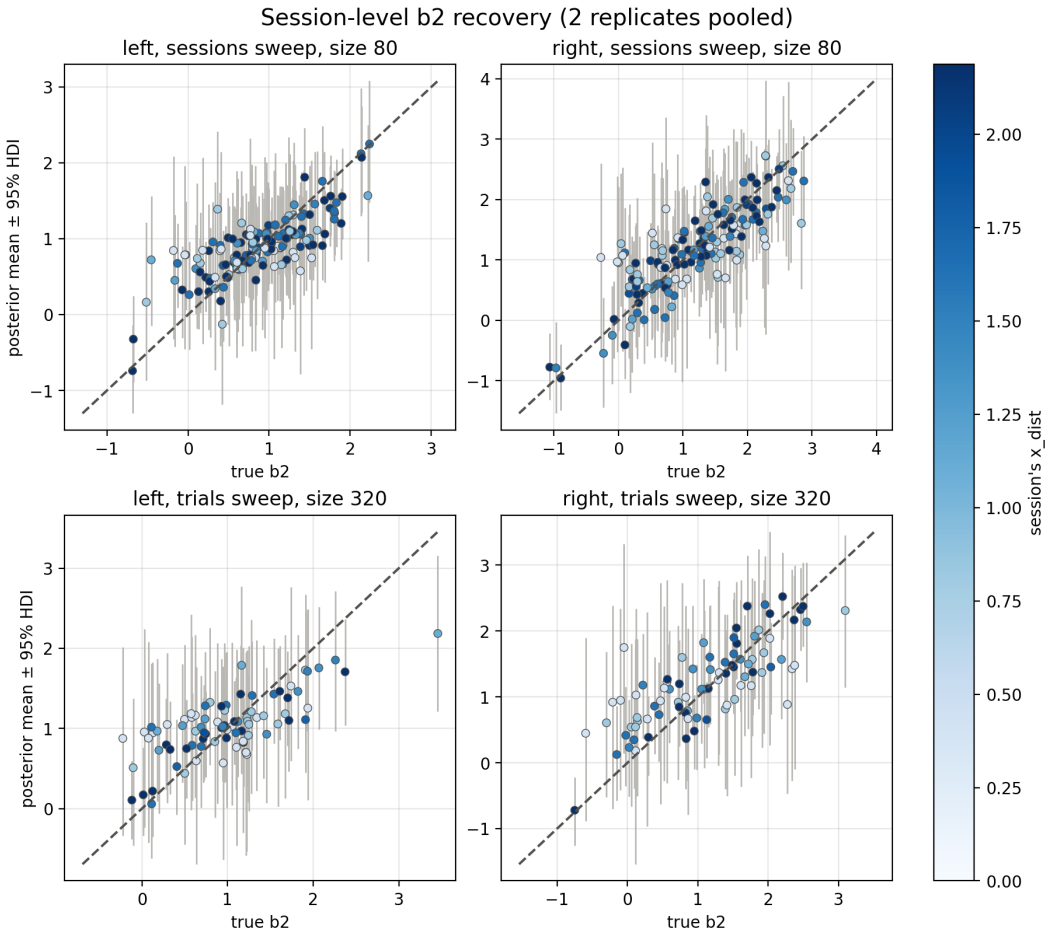


Figure 6: Session-level β_2: posterior mean and 95% HDI against the true value, colored by the session's distractor covariate d_s, at the largest dataset of each sweep (both replicates pooled). The dashed line is the identity.

Table 6: Session-level β_2 by distractor amplitude at 320 trials per session (both hands and replicates pooled; n = number of session estimates).

amplitude (μm)	n	RMSE	mean HDI width	coverage
6	42	0.69	2.58	0.98
12	30	0.44	2.08	1.00
16	6	0.68	1.85	0.83
18	6	0.53	1.69	1.00
20	12	0.37	1.62	1.00
22	6	0.46	1.56	0.83
24	18	0.36	1.43	0.94
26	4	0.54	1.33	0.75
28	2	0.37	1.17	1.00
32	42	0.32	1.19	0.93

Lapse rates γ_h and γ_l. At 320 trials the session lapse rates have RMSE 0.004–0.007 and coverage 0.98–0.99. In the sessions sweep the coverage of γ_h is lower at 40 and 80 sessions (0.86 and 0.87, averaged over hands). With σ_{γ_h} not identified, the session values are pulled strongly toward the hand's typical value, so a miss in that value likely carries over to every session of the hand; at 40 sessions, for example, the right hand's μ_{γ_h} averages -2.60 against a true -2.41.

3.5 Posterior densities

The following plots show the same fits as full posterior densities, darker for larger datasets. They confirm the points above: the means and the sds of the logistic coefficients concentrate around the truth as data grow, while the posteriors of σ_{γ_h} and σ_{γ_l} barely change.

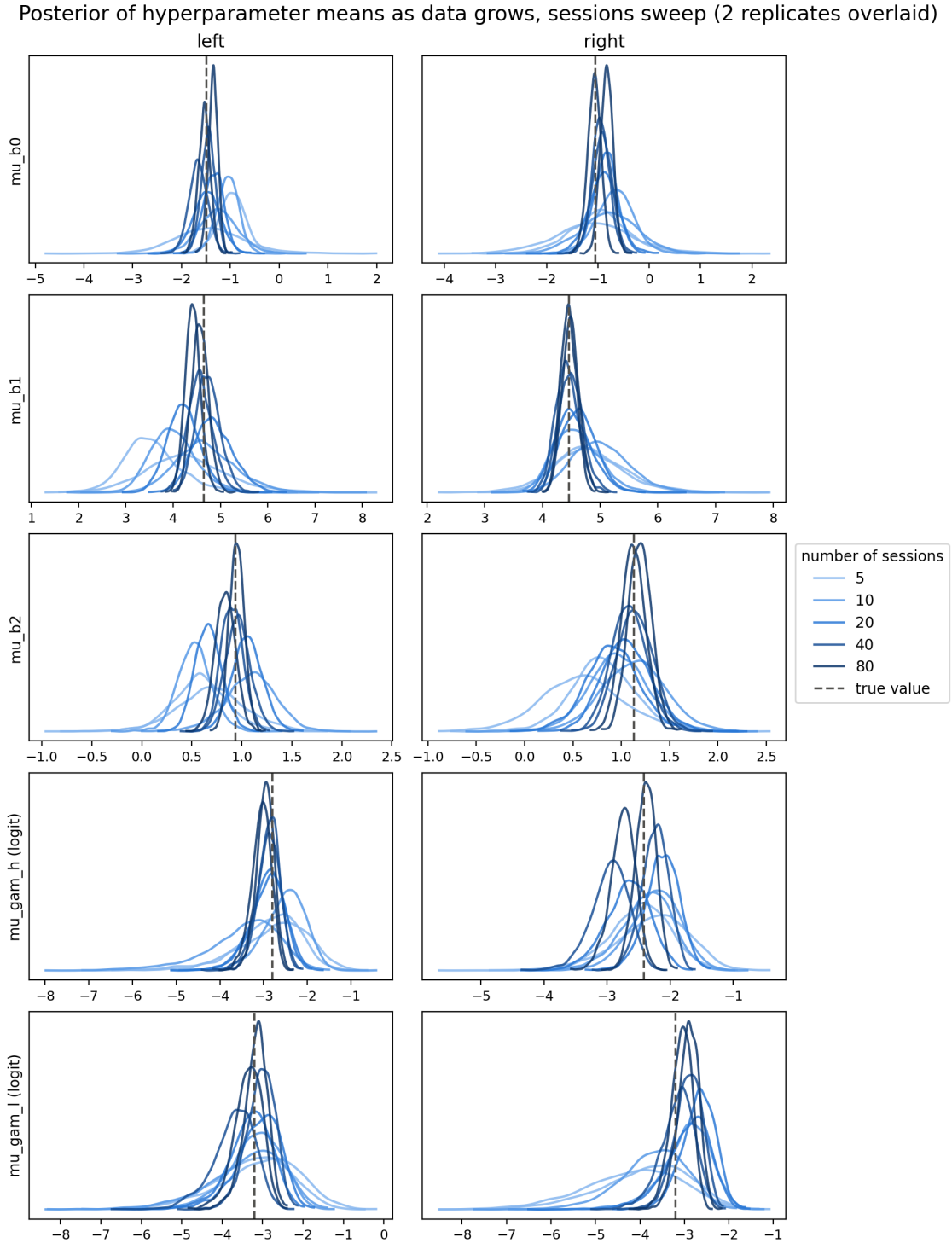


Figure 7: Posteriors of mean hyperparameters as sessions vary

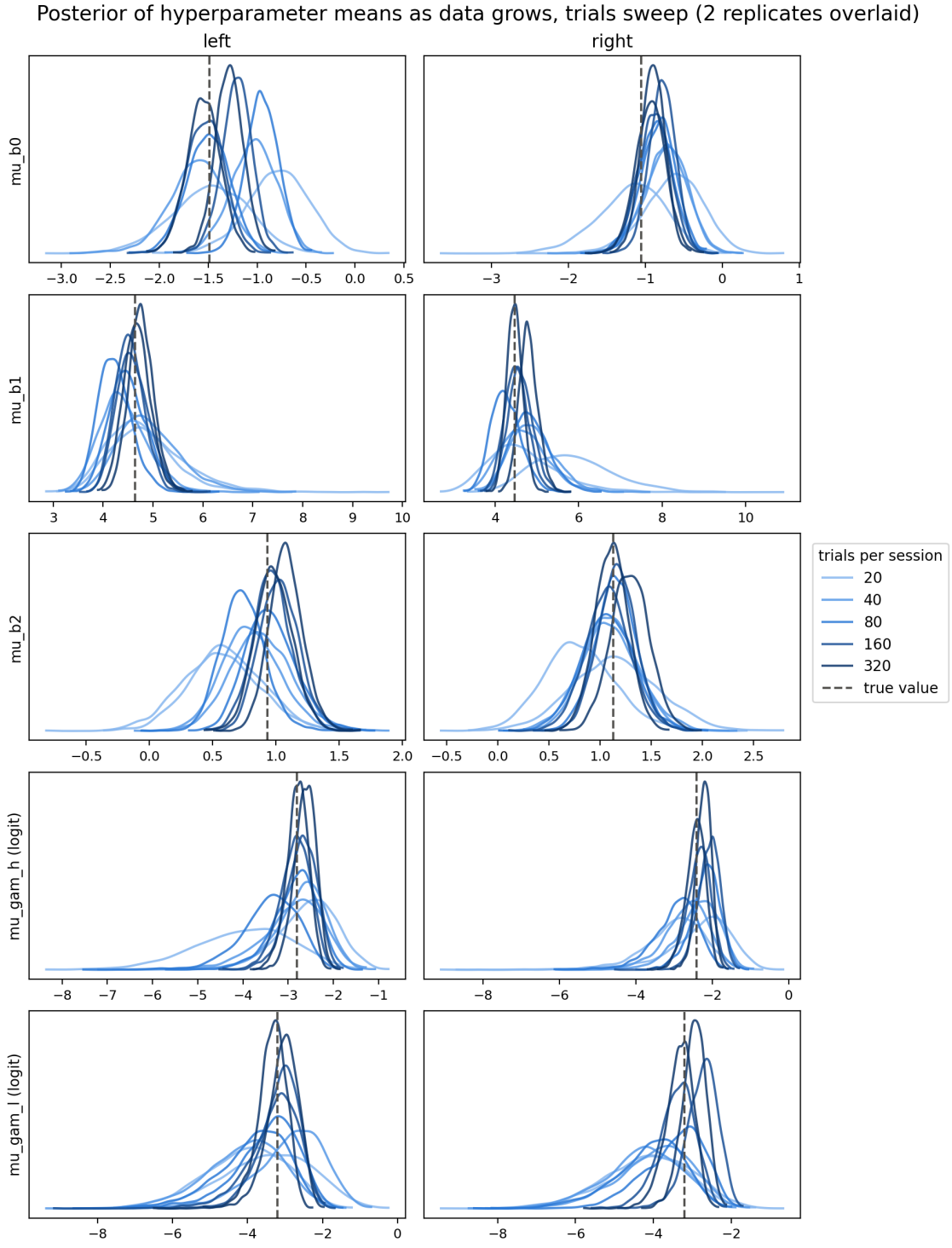


Figure 8: Posteriors of mean hyperparameters as trials vary

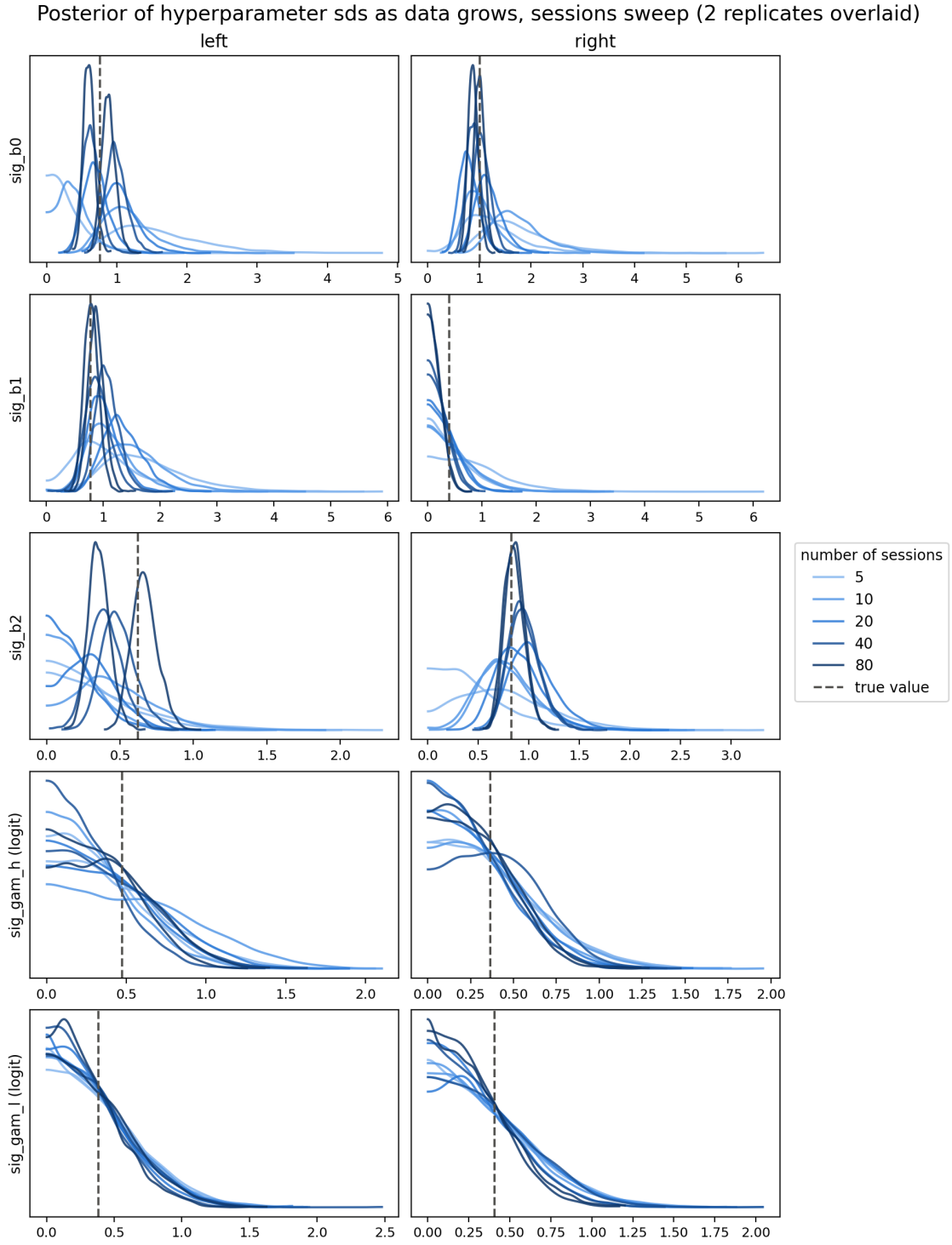


Figure 9: Posteriors of sd hyperparameters as sessions vary

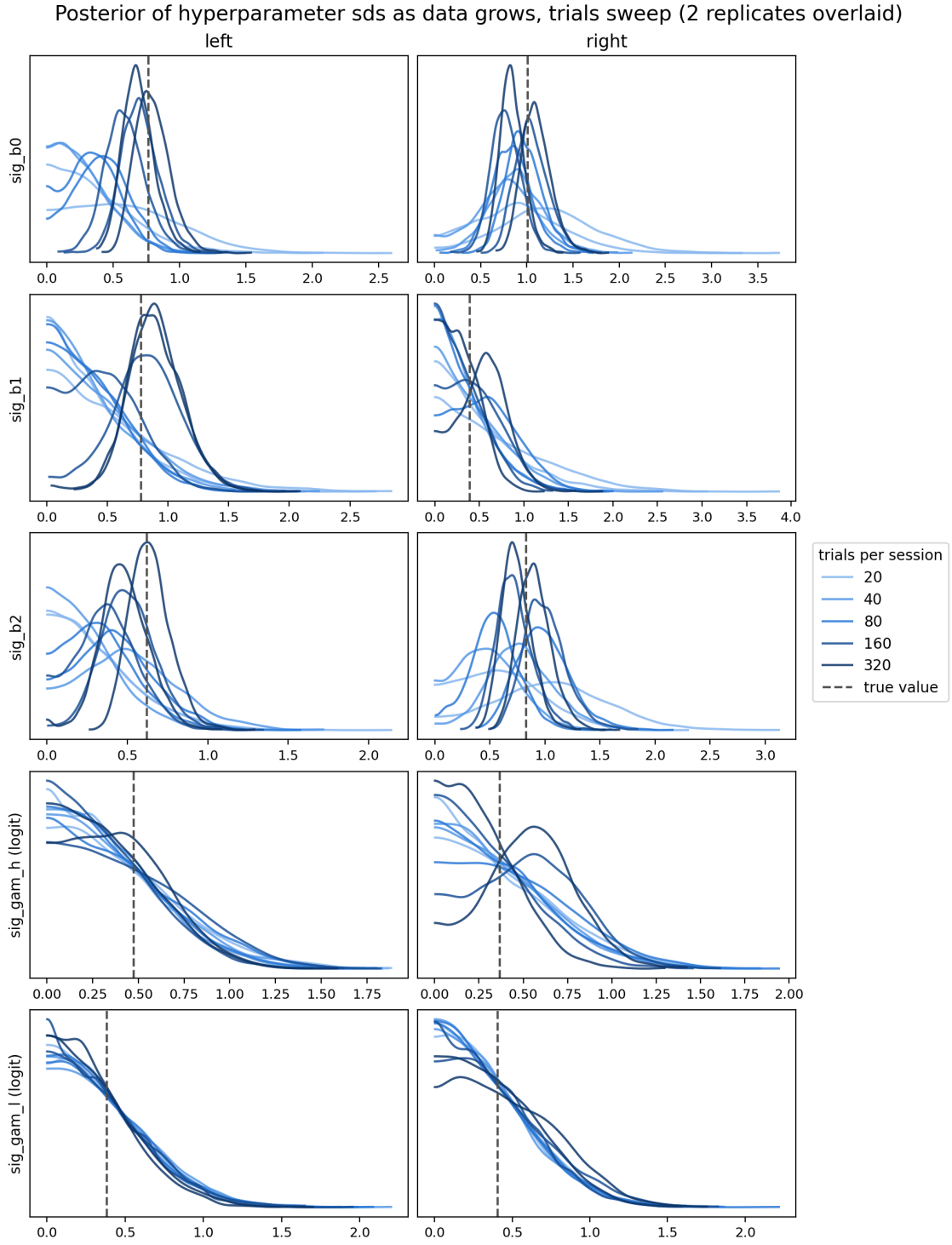


Figure 10: Posteriors of sd hyperparameters as trials vary

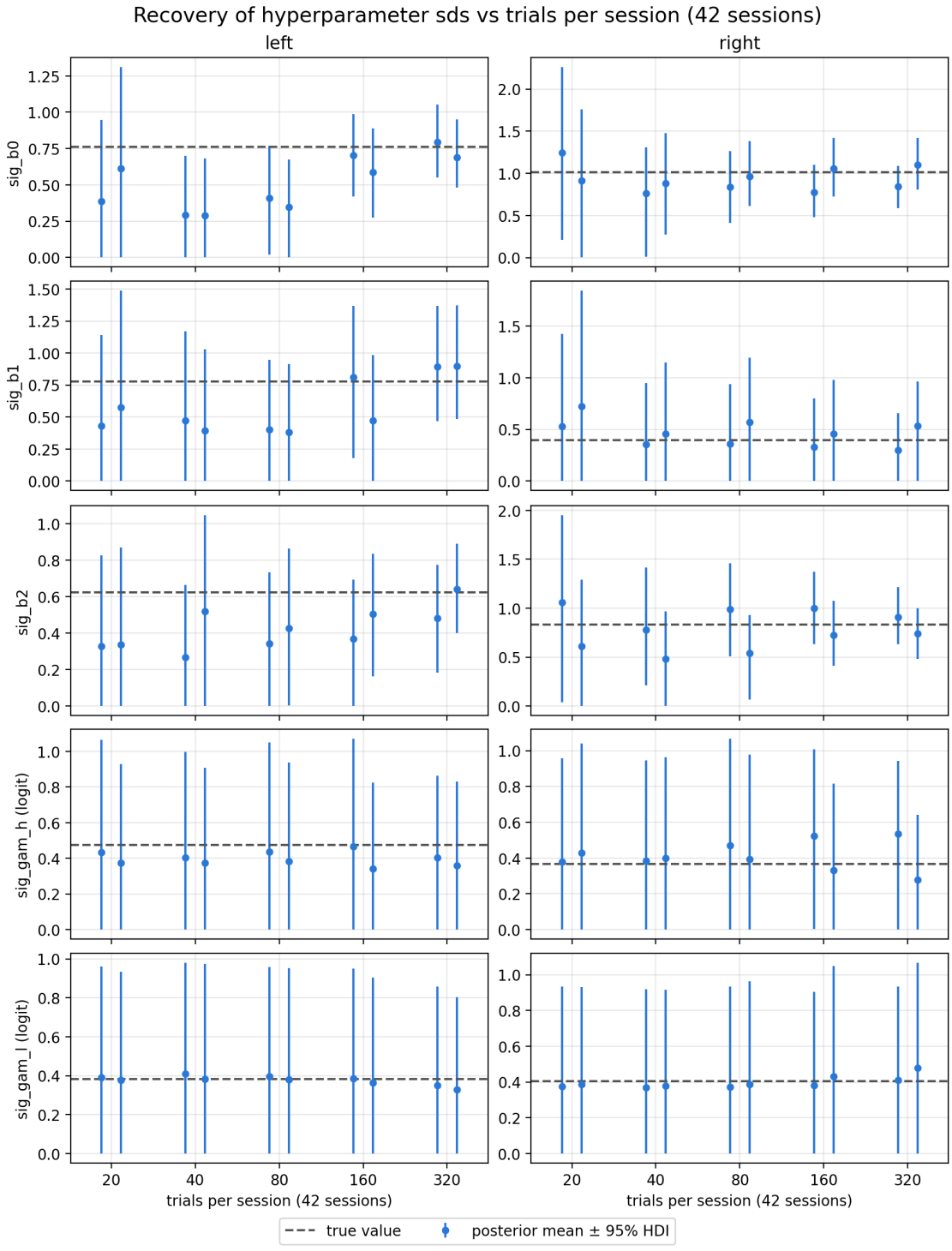


Figure 11: Posterior mean and 95% HDI of the between-session sds against trials per session (42 sessions).

3.6 Conclusions for the real data

At the size of the real dataset (42 sessions of about 308 trials), if the data follow Model C:

- **Reliable:** per-session PSEs in both conditions and JNDs; all five hand-level means, including the distractor effect μ_{β_2}, whose HDI (0.47–0.73 wide at 40 sessions) is narrow enough to tell an effect of the size simulated here from 0; the between-session sds of β_0, β_1 and β_2, with the caveats above.
- **Weakly determined:** the individual session coefficients, especially β_2 for sessions with small distractors, which are pulled toward the hand mean. Session-level β_2 values should not be compared across sessions without their HDIs.
- **Not identified:** σ_{γ_h} and σ_{γ_l}. Their posteriors largely reflect the priors and should not be interpreted.
- The sampler had no real difficulties: 2 divergences in 20 fits, both in the smallest datasets.

What this study does not test. The synthetic data follow Model C exactly. In particular, the unimanual and bimanual trials of a session share their lapse rates, and the distractor effect is linear in amplitude. Model B's fit to the real data suggests that γ_h is higher on bimanual than on unimanual trials, and the synthetic data built from that fit already show more low-stimulus lapses on unimanual trials than the real data do (the comparison printed by `DataProcessing_C.py`). How Model C behaves when these assumptions fail is a question for a fit to the real data and a model comparison with Model B.

Possible follow-ups.

1. Fit Model C to the real data and compare it with Model B by LOO, as was done for Models A and B.
2. Consider lapse rates per hand and condition (unimanual, bimanual), keeping β_2 for the distractor's effect on the curve's position.
3. Consider removing the session-level variation of the lapse rates, since σ_{γ_h} and σ_{γ_l} are not identified (as in Model B).